MONOCHROMATIC DISTANCES IN FINITE COLORINGS OF THE PLANE

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Abstract. We show that every coloring of the plane with finitely many colors has, for all sufficiently large j, two points of the same color at distance 2^j, and two at distance $j!$. This answers questions of Soifer and of Kahle. More generally, no finite coloring of $\mathbb{R}^n$, $n \geq 2$, avoids a set of algebraic distances that is unbounded or accumulates at 0. We also show that the number of colors needed to avoid r distances in the plane grows exponentially in r, with explicit distance sets. This answers Erdős Problem #706 in the negative: every coloring avoiding the distances $1, q, \ldots, q^{K-1}$ uses more than $\frac{22}{15}(2.5^K - 1)$ colors when $q \geq 4096$ is algebraic. Crocker independently derived the qualitative statements conditionally on [20, Theorem 2.3]. The statements for unbounded sets (including those of Soifer and Kahle), a lacunary bound $\frac{4}{9}(3/2)^K$ for integer $q \geq 10^6$ (which already answers #706), and a quantitative Soifer bound are formally verified in Lean 4 with only the standard axioms.

The proofs combine the transfer principle of OpenAI's proof that the plane is not five-colorable, extended to sets of algebraic distances, with explicit Fourier certificates. They also give quantitative bounds. Every k-coloring of the plane has a monochromatic odd distance below $18k^2$, a monochromatic distance $2^i < k^{6.2}$, and a monochromatic distance $s \leq 2X_0 k^{7.44}$ in any algebraic set meeting every $[X, 2X]$, $X \geq X_0$. In $\mathbb{R}^3$ the odd distance can be taken at most k. For two distances, $\chi(\mathbb{R}^2, \{1, d\}) \geq 7$ for every algebraic $d \geq 7$, and $\chi(\mathbb{R}^2, \{1, d\}) \geq 10$ for $d \geq 150$.

1. Introduction

For a set $D \subset (0, \infty)$ let $\chi_D(\mathbb{R}^n) = \chi(\mathbb{R}^n, D)$ be the least number of colors in a coloring of $\mathbb{R}^n$ with no two points of the same color at a distance in D. No regularity is imposed on the color classes. The upper density of a measurable set $A \subset \mathbb{R}^n$ is

$$\bar{\delta}(A) = \limsup_{T \to \infty} \frac{|A \cap [-T, T]^n|}{(2T)^n},$$

taken along origin-centered cubes; this convention is used throughout. Let $m_D(\mathbb{R}^n)$ be the supremum of $\bar{\delta}(A)$ over measurable sets A with no two points at a distance in D, and write $m_1 = m_{\{1\}}$. Let $F_n(r) = \sup_{|D|=r} \chi_D(\mathbb{R}^n)$.

For $D = \{1\}$, $\chi_D(\mathbb{R}^n)$ is the chromatic number of $\mathbb{R}^n$. In the plane OpenAI [20] recently proved $\chi(\mathbb{R}^2) \geq 6$. Its first step is a transfer theorem: a proper coloring of the plane yields a measurable coloring whose monochromatic unit pairs form a null set. In a companion paper [6] we extend the transfer to every dimension and to every countable set of algebraic distances, with a Lean formalization. Crocker [4] independently stated the same extension conditionally on [20, Theorem 2.3]. This paper develops the consequences for colorings that avoid several distances.

Two questions motivate it. First, Erdős [17, 11] asked whether $F_2(r)$ grows polynomially in r; Bukh [2] records this as a conjecture. Croft, Falconer and Guy [7, Problem G11] asked whether it grows exponentially. The best lower bound was $F_2(r) \geq cr\sqrt{\log r}$, from the square grid [7, 19]. Bukh [2] answered the measurable version: $m_D(\mathbb{R}^n) \geq m_1(\mathbb{R}^n)^r$ for every r-element set D, and $m_{t_1 D_1 \cup \cdots \cup t_r D_r}(\mathbb{R}^n) \to \prod_i m_{D_i}(\mathbb{R}^n)$ when all ratios t_i/t_{i+1} tend to infinity. Second, Davies

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proved that every finite coloring of the plane has a monochromatic pair at an odd distance [8], and Davies, McCarty and Pilipczuk did the same for primes and polynomial values [9]. They asked whether powers of 2 (Soifer) and factorials (Kahle) behave likewise [9, Problems 5.4, 5.5]. They also remarked that they knew of no infinite subset of $\mathbb{N}$ that a finite coloring of the plane can avoid. In dimensions $n \geq 4$, Davies [10, Theorems 1.2, 1.3] transferred Bukh's results to arbitrary colorings for distances with rational squares. Since 2^j and $j!$ have rational squares, this settles the questions of Soifer and Kahle in $\mathbb{R}^n$ for $n \geq 4$.

Contributions.

- Theorem 2.1: for countable sets of algebraic distances, $\chi_D(\mathbb{R}^n) \geq 1/m_D(\mathbb{R}^n)$; exponential growth of $F_n(r)$; and no finite coloring of $\mathbb{R}^n$ avoids a set of algebraic distances that is unbounded or accumulates at 0. In particular the questions of Soifer and Kahle have positive answers in $\mathbb{R}^2$ and $\mathbb{R}^3$.
- Section 3: explicit Fourier certificates. Odd distances give monochromatic odd distances below $18k^2$ in the plane and at most k in $\mathbb{R}^3$. Lacunary sets give $F_2(r) \geq 2.5^r$ with explicit distance sets, and polynomial bounds in k for monochromatic powers of 2, factorials and elements of sets meeting every dyadic interval. We also give bounds for two distances and for small explicit sets.
- Section 4: a Lean formalization of the statements for unbounded sets, of a lacunary bound that answers Erdős Problem #706, and of a quantitative Soifer bound.

Relation to [4]. Crocker's note states the transfer for algebraic distance sets, the unavoidability of unbounded sets, and exponential growth for widely separated distances [4, Corollaries 8.2, 8.3], including the application to the questions of Soifer and Kahle. These are conditional on [20, Theorem 2.3]. His distance sets are $\{N, N^2, \dots, N^m\}$ with $N \geq N_0$ for a threshold N_0 that is not explicit. He notes that $\chi(\mathbb{R}^2, \{1, N\}) \geq 17$ for large N. Here the distance sets and the bounds of Section 3 are explicit. The transfer behind Theorem 2.1(1) and the case of Theorem 2.1(3) for sets unbounded above are machine-checked. Since [20, Theorem 2.3] is itself formalized, these are unconditional relative to the Lean kernel. Theorem 2.1(2) rests on Bukh's theorem, which is not formalized. The case of Theorem 2.1(3) for sets accumulating at 0 is not formalized either.

2. Transfer for sets of algebraic distances

Theorem 2.1. *Let $n \geq 2$.*

(1) *Let D be a countable set of positive real algebraic numbers. Every coloring of $\mathbb{R}^n$ with k colors and no monochromatic pair at a distance in D yields k pairwise disjoint measurable sets, each with no two points at a distance in D, that cover $\mathbb{R}^n$ up to a null set. In particular $\chi_D(\mathbb{R}^n) \geq 1/m_D(\mathbb{R}^n)$.*

(2) *For every $r \geq 1$ and $\varepsilon > 0$ there is R such that $\chi_D(\mathbb{R}^n) \geq (m_1(\mathbb{R}^n) + \varepsilon)^{-r}$ for every set $D = \{d_1 < \cdots < d_r\}$ of positive real algebraic numbers with $d_{i+1}/d_i \geq R$. In particular*

$$F_2(r) \;\geq\; m_1(\mathbb{R}^2)^{-r} \;>\; 4.0486^r \qquad \text{and} \qquad F_3(r) \;\geq\; m_1(\mathbb{R}^3)^{-r} \;>\; 6.523^r.$$

(3) *For every coloring of $\mathbb{R}^n$ with finitely many colors there is $R > 0$ such that every positive real algebraic number $s \notin [1/R, R]$ is the distance between two points of the same color. The color may depend on s.*

Proof. (1) This is [6, transfer for sets of algebraic distances] (Lean: `euclidean_transfer_set`), valid for every countable set of positive algebraic distances. The k sets cover $\mathbb{R}^n$ up to a null set, so one of them has upper density at least $1/k$.

(2) Since m_D is scale-invariant, it depends only on the consecutive ratios of D. Suppose that for some ε no R works. Then there are sets D_j with all consecutive ratios at least j and $m_{D_j} > (m_1 + \varepsilon)^r$. This contradicts [2, Theorem 1] with $D_i = \{1\}$ (whose distances are

listed in decreasing order there). Hence there is R with $m_D(\mathbb{R}^n) \le (m_1(\mathbb{R}^n)+\varepsilon)^r$ whenever all consecutive ratios are at least R, and (1) applies. Choosing the d_i rational shows that $F_n(r) \ge (m_1+\varepsilon)^{-r}$ for every $\varepsilon > 0$. The numerical bounds use $m_1(\mathbb{R}^2) \le 0.246998$ [1, Section 8] and $m_1(\mathbb{R}^3) \le 0.1532996$ [3].

(3) A monochromatic distance in a plane $\mathbb{R}^2 \subset \mathbb{R}^n$ is a monochromatic distance in $\mathbb{R}^n$. So we may restrict the coloring to a plane and assume $n = 2$. Let the coloring use k colors, and let B be the set of positive real algebraic numbers that are not monochromatic distances. The set B is countable, and the coloring has no monochromatic pair at a distance in B. By (1) there are measurable sets $A_1, \dots, A_k$, each with no two points at a distance in B, that cover $\mathbb{R}^2$ up to a null set. One of them, say A_1, has positive upper density. By the theorem of Furstenberg, Katznelson and Weiss [18], A_1 realizes every sufficiently large distance, so B is bounded above. The set A_1 also has positive measure. By the Steinhaus theorem, $A_1 - A_1$ contains a neighborhood of 0, so A_1 realizes every sufficiently small distance (see also [12] on small distances in plane sets of positive measure). Hence B is bounded away from 0, and $B \subset [1/R, R]$ for some R. □

The proof of (3) does not use (2), and so it does not depend on Bukh's theorem.

By the de Bruijn–Erdős theorem, $F_2(r)$ is also the supremum of the chromatic numbers of finite graphs in the plane whose edges are the pairs at distances in an r-element set. So (2) answers Erdős Problem #706 of [11], "is it true that $L(r) \le r^{O(1)}$?", in the negative. Davies [10, Theorem 1.2] states Bukh's limit in the same uniform form for $n \ge 4$.

Part (3) is the analogue for arbitrary colorings of the theorem of Furstenberg, Katznelson and Weiss [18]. In the plane it contains, as special cases, that every finite coloring has a monochromatic pair at an odd distance [8] and at a prime distance [9]. It strengthens these to all sufficiently large integers. It also answers the questions of Soifer and of Kahle: every finite coloring of the plane has a monochromatic pair at distance 2^j, and one at distance $j!$, for all sufficiently large j. By (3) there is no infinite subset of $\mathbb{N}$ that a finite coloring of the plane can avoid. The same holds for infinite sets of algebraic numbers that are unbounded or accumulate at 0.

3. Fourier certificates and quantitative bounds

3.1. **The Delsarte framework in $\mathbb{R}^n$.** For $t > 0$ let

$$\Omega_n(t) = \Gamma(n/2)\,(2/t)^{(n-2)/2} J_{(n-2)/2}(t),$$

and let $\Omega_n(0) = 1$. This is the Fourier transform of the normalized surface measure on the unit sphere of $\mathbb{R}^n$, evaluated at a frequency of norm t. In particular $\Omega_2 = J_0$ and $\Omega_3(x) = \sin x/x$. We use the following facts:

- $J_0 \ge -\rho$ with $\rho = 0.40276$, and $J_0(x) \ge 1 - x^2/4$; also $|J_0(x)| \le \sqrt{2/(\pi x)}$ for $x > 0$, see [15, §13.74];
- $\Omega_3 \ge -0.21724$, $\Omega_3(x) \ge 1 - x^2/6$, and $|\Omega_3(x)| \le 1/x$ for $x > 0$.

Lemma 3.1. *Let $D \subset (0,\infty)$ be finite, and let v_d $(d \in D)$ be real numbers with*

$$h(t) \;=\; 1 + \sum_{d\in D} v_d\, \Omega_n(dt) \;\ge\; 0 \qquad (t \ge 0). \tag{1}$$

Then $m_D(\mathbb{R}^n) \le 1/(1+\sum_d v_d)$. If moreover every $d \in D$ is algebraic, then $\chi_D(\mathbb{R}^n) \ge 1+\sum_d v_d$.

Proof. The first claim is due to de Oliveira Filho and Vallentin [21]; we recall the argument. Since D is bounded, m_D is the supremum of the densities of periodic D-avoiding sets. To see this, truncate to a large cube, remove a boundary layer of width $\max D$, and repeat periodically, as in [4, Lemma 2.2]. Let A be a periodic D-avoiding set of density δ. The spherical average of

its autocorrelation is $f(s) = \int \Omega_n(st)\, d\mu(t)$ for a positive measure μ of mass δ with $\mu(\{0\}) = \delta^2$, and $f(d) = 0$ for $d \in D$. Hence $\delta = \int h\, d\mu \geq h(0)\delta^2$, so $\delta \leq 1/h(0)$. The second claim follows from Theorem 2.1(1). □

3.2. **Odd distances.**

Theorem 3.2. *For every $N \geq 1$, $\chi(\mathbb{R}^2, \{1, 3, 5, \ldots, 2N-1\}) \geq 1+N/(2.5\sqrt{N}+1)$. Consequently every coloring of the plane with k colors has two points of the same color at an odd distance less than $18k^2$.*

Proof. Let $K_N(x) = \sum_{|j|<N}(1 - |j|/N)e^{ijx} = \sin^2(Nx/2)/(N\sin^2(x/2))$ be the Fejér kernel, and $P(u) = \cos u\, K_N(2u)$. Expanding the product shows $P(u) = \sum_{j=0}^{N-1} c_j \cos((2j+1)u)$ with $c_j = 2(1 - \frac{2j+1}{2N}) \geq 0$ and $\sum_j c_j = P(0) = N$. By $J_0(x) = \frac{1}{\pi}\int_0^\pi \cos(x\sin\theta)\, d\theta$,

$$G(t) := \sum_j c_j J_0((2j+1)t) = \frac{2}{\pi}\int_0^{\pi/2} P(t\sin\theta)\, d\theta \;\geq\; -\frac{2}{\pi}\int_{\{u\in[0,t]:\cos u<0\}} \frac{K_N(2u)\, du}{\sqrt{t^2-u^2}}.$$

The integral is empty for $t \leq \pi/2$. Use $K_N(2u) \leq \min(N, \pi^2/(4Ns^2))$ with $s = \operatorname{dist}(u, \pi\mathbb{Z})$, $\int K_N(2u)\, du = \pi$ over each period, and $(t^2-u^2)^{-1/2} \leq (t(t-u))^{-1/2}$. The peaks of $K_N(2u)$ in $\{\cos u < 0\}$ sit at odd multiples of π, which are 2π apart. A peak u_ℓ with $t - u_\ell \geq \pi$ contributes at most $\pi\sqrt{2}/\sqrt{t(t-u_\ell)}$. These values of $t - u_\ell$ are at least π and 2π apart, so together they contribute at most $\sqrt{2\pi/t}\,\sum_{m\leq t/(2\pi)}(2m-1)^{-1/2} \leq \sqrt{2}$. At most one peak lies in $(t-\pi, t+\pi/2)$, since the peaks are 2π apart. Its contribution is bounded by the Hardy–Littlewood rearrangement inequality. The one-sided kernel $y^{-1/2}\mathbf{1}_{y>0}$ has symmetric decreasing rearrangement $(2|y|)^{-1/2}$, so the bound is $t^{-1/2}\int_{\mathbb{R}} \min(N, \pi^2/(4Ns^2))(2|s|)^{-1/2} ds = \frac{8}{3}\sqrt{\pi}\sqrt{N}\, t^{-1/2}$, with $t \geq \pi/2$. Altogether $G(t) \geq -\frac{2}{\pi}\big(\frac{8}{3}\sqrt{\pi}\sqrt{2/\pi}\,\sqrt{N} + \sqrt{2}\big) \geq -(2.5\sqrt{N}+1)$. So (1) holds with $n = 2$ and $v_{2j+1} = c_j/(2.5\sqrt{N}+1)$, and Lemma 3.1 gives the first claim. For the second, take $N = 9k^2$, for which $N/(2.5\sqrt{N}+1) \geq k$. □

For measurable colorings the order $\sqrt{N}$ is implicit in Steinhardt's spectral argument [14]. For arbitrary colorings, the proof in [8] that the odd-distance graph has infinite chromatic number gives no explicit bound on the odd distance. In $\mathbb{R}^3$ the bound becomes linear.

Theorem 3.3. *For every $N \geq 1$, $\chi(\mathbb{R}^3, \{1, 3, \ldots, 2N-1\}) \geq 1 + 2N$. Consequently every coloring of $\mathbb{R}^3$ with k colors has two points of the same color at an odd distance at most k.*

Proof. With c_j and $P(u) = \cos u\, K_N(2u)$ as in the proof of Theorem 3.2,

$$G_3(t) := \sum_j c_j\, \Omega_3((2j+1)t) = \frac{1}{t}\int_0^t P(u)\, du = \frac{1}{t}\sum_j c_j \frac{\sin((2j+1)t)}{2j+1} = \frac{\pi}{2t}\,\sigma(t),$$

where σ is the Fejér mean of order $2N$ of the square wave $\operatorname{sgn}(\sin t)$, since $c_j = 2(1 - \frac{2j+1}{2N})$ for $j < N$. Hence $|\sigma| \leq 1$. On $[0, \pi]$, $\sigma \geq 0$. Indeed, the partial sums $\sum_{j\leq m}\sin((2j+1)t)/(2j+1) = \int_0^t \frac{\sin(2(m+1)u)}{2\sin u}\, du$ are positive on $(0, \pi)$ (Fejér; see [16, §4.1]), and Abel summation with the decreasing coefficients c_j preserves this. So $G_3(t) \geq 0$ for $t \leq \pi$ and $G_3(t) \geq -\pi/(2t) \geq -\frac{1}{2}$ for $t \geq \pi$. The certificate $v_{2j+1} = 2c_j$ satisfies (1) with $n = 3$ and $\sum v = 2N$, and Lemma 3.1 gives the first claim. For the second, take $N = \lceil k/2 \rceil$. □

By the theorem of Graham, Rothschild and Straus [13], at most four points of $\mathbb{R}^3$ have pairwise odd distances. So the bound of Theorem 3.3 is not a clique bound.

3.3. **Lacunary sets.**

Theorem 3.4. *Let (n, q, λ, c) be a row of Table 1, let $K \geq 1$, and let $d_0 < \cdots < d_{K-1}$ be positive real algebraic numbers with $d_{j+1}/d_j \geq q$ for all j. Then*

$$\chi\big(\mathbb{R}^n, \{d_0, \ldots, d_{K-1}\}\big) \;\geq\; 1 + c\big(\lambda^K - 1\big).$$

dimension n	ratio q	base λ	constant c
2	4096	2.5	22/15
2	32	1.75	5/3
3	256	4	4/3
3	8	2.24	1.6

TABLE 1. Parameter sets for Theorem 3.4.

Proof. Write $\Omega = \Omega_n$, and let ρ_n, β_n and $\gamma_n(x)$ be the constants and envelope from the list of facts: $\rho_2 = 0.40276$, $\beta_2 = \frac{1}{4}$, $\gamma_2(x) = \sqrt{2/(\pi x)}$, and $\rho_3 = 0.21724$, $\beta_3 = \frac{1}{6}$, $\gamma_3(x) = 1/x$. So $\Omega \geq -\rho_n$, $\Omega(x) \geq 1 - \beta_n x^2$, and $|\Omega(x)| \leq \gamma_n(x)$. Take $v_j = v\lambda^j$ and a threshold $A > 0$, with v and A given in Table 2. Then $\sum_j v_j = \frac{v}{\lambda - 1}(\lambda^K - 1)$, and $\frac{v}{\lambda-1} \geq c$ in each row.

Fix $t > 0$ and let m be the largest index $j \leq K-1$ with $d_j t \leq A$, or $m = -1$ if there is none. For $j \leq m$ we have $d_j t \leq A q^{j-m}$. For $j > m$ we have $d_j t > A q^{j-m-1}$. The terms with $j \leq m$ contribute at least

$$\sum_{j \leq m} v_j - \beta_n A^2 \sum_{j \leq m} v_j q^{2(j-m)} \;\geq\; \sum_{j \leq m} v_j - \beta_n A^2 L v_m, \qquad L = \Big(1 - \frac{1}{\lambda q^2}\Big)^{-1}.$$

The term $j = m+1$ contributes at least $-\rho_n v_{m+1}$. The terms $j \geq m+2$ contribute at least $-\sum_{i \geq 1} v_{m+1+i}\gamma_n(Aq^i) = -\kappa v_{m+1}$, where

$$\kappa = \sqrt{\frac{2}{\pi A}}\,\frac{\lambda/\sqrt{q}}{1 - \lambda/\sqrt{q}} \quad (n = 2), \qquad \kappa = \frac{1}{A}\,\frac{\lambda/q}{1 - \lambda/q} \quad (n = 3).$$

Since $\sum_{j \leq m} v_j = v_m \lambda(1 - \lambda^{-m-1})/(\lambda - 1)$ and $v_{m+1} = \lambda v_m$, we get $h(t) = 1 + \sum_j v_j \Omega(d_j t) \geq 1 + v_m b_m$ for $m \geq 0$, where

$$b_m = \frac{\lambda(1 - \lambda^{-m-1})}{\lambda - 1} - \beta_n A^2 L - \rho_n \lambda - \kappa\lambda.$$

For $m = -1$ we get $h(t) \geq 1 - v(\rho_n + \kappa)$. The first term of b_m is 1 for $m = 0$ and at least $(\lambda+1)/\lambda$ for $m \geq 1$. Table 2 lists upper bounds for $\beta_n A^2 L$, $\rho_n\lambda$ and $\kappa\lambda$. In each row $b_m > 0$ for $m \geq 1$, $1 + v b_0 > 0$, and $1 - v(\rho_n + \kappa) > 0$. So (1) holds, and Lemma 3.1 gives the bound. □

n	q	λ	v	A	$\kappa \leq$	$\beta_n A^2 L \leq$	$\rho_n\lambda \leq$	$\kappa\lambda \leq$	$(\lambda+1)/\lambda \geq$
2	4096	2.5	11/5	1/2	0.046	0.063	1.007	0.115	1.4
2	32	1.75	5/4	0.83	0.3923	0.1724	0.7049	0.6866	1.5714
3	256	4	4	1/2	0.0318	0.0417	0.8690	0.1272	1.25
3	8	2.24	1.99	1.38	0.2818	0.3197	0.4867	0.6313	1.4464

TABLE 2. Certificate parameters for the proof of Theorem 3.4.

For example, in the first row $b_m \geq 1.4 - 0.063 - 1.007 - 0.115 > 0$ for $m \geq 1$, $b_0 > -0.19$ with $1 - \frac{11}{5} \cdot 0.19 > 0$, and $1 - \frac{11}{5}(0.40276 + 0.046) > 0$. In the second row the three checks are $1.5714 - 0.1724 - 0.7049 - 0.6866 > 0$, $1 + \frac{5}{4}(1 - 0.1724 - 0.7049 - 0.6866) > 0$ and $1 - \frac{5}{4}(0.40276 + 0.3923) > 0$. In the fourth row they are $1.4464 - 0.3197 - 0.4867 - 0.6313 > 0$, $1 + 1.99(1 - 0.3197 - 0.4867 - 0.6313) > 0$ and $1 - 1.99(0.21724 + 0.2818) > 0$. In the third row $b_m \geq 0.21$ for $m \geq 1$, $b_0 \geq -0.04$, and $1 - 4(0.21724 + 0.0318) > 0$.

Taking $d_j = 4096^j$ in the first row gives $F_2(r) \geq 1 + \frac{22}{15}(2.5^r - 1) \geq 2.5^r$ with the explicit distance sets $\{1, 4096, \ldots, 4096^{r-1}\}$. The same certificates show that measurable sets in $\mathbb{R}^n$ avoiding such distances have density at most $(1 + c(\lambda^K - 1))^{-1}$, for any real d_j. By Bukh's theorem the optimal base for widely separated distances in the plane is $1/m_1(\mathbb{R}^2) > 4.04$. That version gives no explicit ratios.

Corollary 3.5. *Every coloring of the plane with $k \geq 2$ colors has two points of the same color at a distance $2^i < k^{6.2}$. Every coloring of $\mathbb{R}^3$ with $k \geq 2$ colors has two points of the same color at a distance $2^i < k^{2.58}$.*

Proof. In the plane, apply the second row of Table 1 to $d_j = 32^j$, $j < K$, with $K = \lceil \log_{1.75} k \rceil$. Then $1 + \frac{5}{3}(1.75^K - 1) > k$, so some 32^j with $j \leq K - 1$ is a monochromatic distance, and $32^{K-1} < k^{5/\log_2 1.75} < k^{6.2}$. In $\mathbb{R}^3$, apply the fourth row to $d_j = 8^j$ with $K = \lceil \log_{2.24} k \rceil$. Then $1 + 1.6(2.24^K - 1) > k$ and $8^{K-1} < k^{3/\log_2 2.24} < k^{2.58}$. □

Corollary 3.6. *Every coloring of the plane with $k \geq 2$ colors has two points of the same color at distance $j!$ for some $j \leq 31 + \log_{1.75} k$.*

Proof. Apply the second row of Table 1 to $d_i = (31 + i)!$, $i < K$, with $K = \lceil \log_{1.75} k \rceil$. The ratios are $d_{i+1}/d_i = 32 + i \geq 32$. Some $(31 + i)!$ with $i \leq K - 1$ is a monochromatic distance, and $31 + i \leq 30 + K \leq 31 + \log_{1.75} k$. □

Corollary 3.7. *Let S be a set of positive real algebraic numbers that meets every interval $[X, 2X]$ with $X \geq X_0$. Then every coloring of the plane with $k \geq 2$ colors has two points of the same color at a distance $s \in S$ with $s \leq 2X_0 k^{7.44}$. For example, the primes, the integers and the powers of 2 satisfy the hypothesis with $X_0 = 1$, and the perfect squares with $X_0 = 9/2$. For the squares the bound is $9k^{7.44}$.*

Proof. Choose $s_0 \in S \cap [X_0, 2X_0]$ and $s_{j+1} \in S \cap [32s_j, 64s_j]$. With $K = \lceil \log_{1.75} k \rceil$, the distances $s_0, \ldots, s_{K-1}$ have consecutive ratios at least 32. By the second row of Table 1, some s_j with $j \leq K - 1$ is a monochromatic distance, and $s_{K-1} \leq 2X_0 \cdot 64^{K-1} < 2X_0 k^{6/\log_2 1.75} < 2X_0 k^{7.44}$.

For the examples: every real $X \geq 1$ has a prime in $(X, 2X]$ (Bertrand's postulate), an integer in $[X, 2X]$, and a power of 2 in $[X, 2X]$. For the squares, let $X \geq 9/2$ and $m = \lceil \sqrt{X} \rceil$. If $X \leq (1 + \sqrt{2})^2$ then $m \leq 3$ and $m^2 \leq 9 \leq 2X$. Otherwise $m^2 < (\sqrt{X} + 1)^2 \leq 2X$. The value $9/2$ cannot be lowered much: $[4.2, 8.4]$ contains no square. For the positive values of a nonconstant integer-valued polynomial with positive leading coefficient the hypothesis holds for some X_0 depending on the polynomial. □

This makes the theorems of [8, 9] on odd, prime and polynomial distances effective, with polynomial bounds. For odd distances Theorem 3.2 gives the better bound $18k^2$.

3.4. **Two distances.** For two distances a direct two-term certificate does better than Theorem 3.4.

Proposition 3.8. *Let d be a real algebraic number.*

(1) *If $d \geq 10^4$, then $\chi(\mathbb{R}^2, \{1, d\}) \geq 12$ and $\chi(\mathbb{R}^3, \{1, d\}) \geq 31$. If $d \geq 10^{10}$, then $\chi(\mathbb{R}^2, \{1, d\}) \geq 13$.*

(2) *If* $d \in [1.77, 1.92] \cup [7, \infty)$*, then* $\chi(\mathbb{R}^2, \{1, d\}) \geq 7$*. If* $d \geq 16$*, then* $\chi(\mathbb{R}^2, \{1, d\}) \geq 8$*. If* $d \geq 40$*, then* $\chi(\mathbb{R}^2, \{1, d\}) \geq 9$*. If* $d \geq 150$*, then* $\chi(\mathbb{R}^2, \{1, d\}) \geq 10$*.*

Proof. (1) Take $h(t) = 1 + aJ_0(t) + bJ_0(dt)$ and $A > 0$. If $t > A$, then $h \geq 1 - \rho a - b\sqrt{2/(\pi dA)}$. If $t \leq A < dt$, then $h \geq 1 + a(1 - A^2/4) - \rho b$. If $dt \leq A$, both Bessel values are positive. With $\rho = 0.40276$, $A = 0.3$, $a = 2.1976$, $b = 7.8131$ the first two bounds are at least 0.0011 and 0.0013 for all $d \geq 10^4$, and $1 + a + b > 11$. With $A = 0.05$, $a = 2.4794$, $b = 8.6317$ they are at least 0.0011 and 0.0013 for $d \geq 10^{10}$, and $1 + a + b > 12$. In $\mathbb{R}^3$, use Ω_3, $\rho_3 = 0.21724$, $1 - A^2/6$ and the envelope $1/(dA)$, with $A = 0.3$, $a = 4.5595$, $b = 25.2703$. The bounds are at least 0.0010 and 0.0013, and $1 + a + b > 30$. Lemma 3.1 applies in each case.

(2) For $d \geq 10^4$ this follows from (1). For $d \in [d_1, d_2] \subset [7, 10^4]$ we use $h(t) = 1 + aJ_0(t) + bJ_0(dt)$ with $b \geq 0$, one pair (a, b) for all d in the interval. We bound $J_0(dt)$ below by $\min\{J_0(x) : d_1 t \leq x \leq d_2 t\}$. This minimum is computed from a tabulation of J_0 on $[0, 2050]$ with spacing 10^{-4} and a grid error term, and replaced by $-\sqrt{2/(\pi d_1 t)}$ when $d_1 t > 2000$. Positivity is checked on a t-grid with a Lipschitz margin, and beyond a cutoff T by the envelope bound. Thirteen intervals cover $[7, 10^4]$ and two cover $[1.77, 1.92]$, each with $1 + a + b > 6$. For the bounds 8, 9 and 10 we use 12, 11 and 10 intervals covering $[16, 10^4]$, $[40, 10^4]$ and $[150, 10^4]$, together with (1). The intervals and coefficients are in the accompanying files. All covers in this proposition were re-verified independently with ball arithmetic (Arb), including the exact decimal endpoints, with margins at least 0.01. □

For $d \in (1.93, 7)$ the Delsarte bound for $m_{\{1,d\}}(\mathbb{R}^2)$ exceeds $1/6$, and we could not close this range with inclusion–exclusion constraints. Inside it, $\chi(\mathbb{R}^2, \{1, (\sqrt{2}+\sqrt{6})/2\}) \geq 7$ holds by finite certificates [5].

Theorem 2.1(2) with $r = 2$ gives $\chi(\mathbb{R}^2, \{1, d\}) \geq 17$ and $\chi(\mathbb{R}^3, \{1, d\}) \geq 43$ for every sufficiently large algebraic d, since $m_1(\mathbb{R}^2)^{-2} \geq 1/0.246998^2 > 16.39$ and $m_1(\mathbb{R}^3)^{-2} \geq 1/0.1532996^2 > 42.55$. The threshold for d is not explicit. Crocker [4] notes the planar bound 17 for large N. The bounds 12, 13 and 31 of Proposition 3.8 are versions with explicit thresholds.

3.5. **Small explicit sets.** Numerically, the optimal certificates in (1) give about $2.9\sqrt{N}$ for odd distances, about 1.3^K for $D = \{1, 2, 4, \ldots, 2^{K-1}\}$, and faster than linear growth for $D = \{1, 4, 9, \ldots, N^2\}$ in the ranges computed ($K \leq 8$, $N \leq 16$). We do not know whether the growth for powers of 2 persists, since the high-frequency terms of lacunary sums do not cancel uniformly. Rounded and shrunk rational solutions certify the following.

Proposition 3.9. *Every coloring of the plane with* k *colors has two points of the same color at a distance in* D*, for*

$$\begin{aligned} k = 6: &\quad D = \{1, 3, 5, 7\};\\ k = 8: &\quad D = \{1, 3, 5, \ldots, 15\};\\ k = 10: &\quad D = \{2, 3, 5, 7\};\\ k = 16: &\quad D = \{1, 2, 4, 8, 16, 32\};\\ k = 18: &\quad D = \{1, 4, 9, \ldots, 64\}. \end{aligned}$$

Proof. For each set we give a certificate (v_d) satisfying (1) with $n = 2$ and $1 + \sum_d v_d > k$, and apply Lemma 3.1. Each certificate is checked in floating point, not in ball arithmetic. The check uses a grid with margin 0.03 against a Lipschitz error of at most 0.005, together with the bound $|J_0(x)| \leq \sqrt{2/(\pi x)}$ beyond the grid. □

4. Formal verification

The Lean development builds on the library of [20] and on the formalization of the transfer in [6] (`OAI.HNN.euclidean_transfer_set`). The directory `OAI/Soifer` has 1431 lines. It contains the following.

- A weak form of the Furstenberg–Katznelson–Weiss theorem, proved by Parseval on a torus of side $3T$ and the circle decay bound of [20]: a measurable set in the plane with density at least $c > 0$ in arbitrarily large origin-centered squares realizes some distance of every set of positive reals that is unbounded above. Combined with `euclidean_transfer_set` this gives `mono_distance_of_unbounded`: for every finite coloring of the plane and every set S of positive algebraic reals that is unbounded above, some $s \in S$ is a monochromatic distance. This is Theorem 2.1(3) for sets unbounded above. It gives `soifer` and `kahle`:
  ```
  theorem soifer {k : Nat} [NeZero k] (c : EuclideanSpace R (Fin 2) -> Fin k) :
      exists N : Nat, forall n >= N, exists x y, dist x y = (2:R)^n /\ c x = c y
  ```
  and the same with $n!$ in place of 2^n. The statements `soifer_euclid` and `kahle_euclid` give the same in $\mathbb{R}^n$ for every $n \geq 2$, by restriction to a coordinate plane. For $n \geq 4$ they already follow from [10, Theorem 1.3], so they are new only for $n = 2, 3$.
- A torus version of the Delsarte bound (`torus_delsarte`), and a lacunary certificate built from $|\hat{\sigma}| \leq 1$, $\operatorname{Re}\hat{\sigma}(\xi) \geq 1 - 2\pi^2|\xi|^2$ and the decay bound of [20], proved by induction on K. Together they give `lacunary_chromatic`: every coloring of the plane avoiding the distances $(10^6)^j$, $j < K$, has at least $\frac{4}{9}(3/2)^K$ colors. The statement `lacunary_chromatic_q` is the same for the distances q^j, $j < K$, for every integer $q \geq 10^6$.
- `erdos706_not_polynomial`: for all C and e there is a K-element set of positive algebraic distances that no coloring of the plane with at most CK^e colors avoids. The set used is $\{(10^6)^j : j < K\}$, which consists of integers. By compactness, `erdos706_finite` gives the same for the distance graph on a finite point set $P \subset \mathbb{R}^2$, as in Erdős's formulation. This is a formal negative answer to Erdős Problem #706.
- `soifer_quantitative`: from `lacunary_chromatic_q` with $q = 2^{20}$, every coloring of the plane with k colors has two points of the same color at a distance 2^i with $i \leq 20(2\lfloor \log_2 9k \rfloor + 2)$. This is a formal version, with a weaker exponent, of the planar part of Corollary 3.5.

`#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound` for every statement listed. The following are not formalized: the Fourier certificates of Section 3, Bukh's theorem used in Theorem 2.1(2), and the case of Theorem 2.1(3) for sets accumulating at 0.

5. Remarks

Algebraic versus transcendental distances. Bukh [2] wrote that it is very likely that the ordinary chromatic number does not share the exponential growth of the measurable one. He conjectured that $\chi_D(\mathbb{R}^n)$ is bounded when the elements of D are algebraically independent. Davies, McCarty and Pilipczuk [9, Conjecture 5.6] record the stronger form that every algebraically independent set (in particular an unbounded one) can be avoided by a finite coloring of the plane. Theorem 2.1 shows that for algebraic distances the exponential growth does transfer. As far as we can see, the only step of the transfer that uses algebraicity is the number-field argument in the proof of [20, Theorem 2.3] (Lemma 3.4 and §3.5 there). The other steps appear to work over any countable real closed subfield $F \subset \mathbb{R}$ with $E = F(i)$. Suppose the rigidity theorem held for $E = F(i)$ with F the real closure of $\mathbb{Q}(S)$, for a countable unbounded algebraically independent set S. Then the argument of Theorem 2.1(3) would show that no finite coloring of the plane avoids S, contradicting that conjecture. So either the rigidity theorem is genuinely

arithmetic, or the conjecture fails. In dimension one the transfer fails already for $D = \{1, \sqrt{2}\}$: a Hamel-basis parity coloring avoids D with two colors, while no measurable two-coloring does.

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